Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/12/2006
PI Name:	Sytkowski, Arthur J. M.D.		
Project Title:	Erythroid Cell Growth And Differentiation In Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	asytkows@caregroup.harvard.edu	**Fax:**	FY 617-632-8005
PI Organization Type:	UNIVERSITY	**Phone:**	617-632-9980
Organization Name:	Beth Israel Deaconess Medical Center/Harvard Medical Center		
PI Address 1:	330 Brookline Ave., W/BL 548		
PI Address 2:			
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02215	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	10/31/2005
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1592		
Performance Goal No.:			
Performance Goal Text:			

The long-term objective of this research is the study of erythroid cell growth and differentiation in space. We propose to study the effects of microgravity on erythroid cell surface receptors and their interaction with their cognate ligand; receptor-initiated signal transduction; and on the assembly of the cytoskeleton. The work will be done initially through a ground-based experimental approach using the simulated microgravity environment of the NASA-Rotating Wall Vessel bioreactor (RWV), which has already been proven useful in studies of normal and malignant human cell growth and development. This research will serve as a basis for later experiments to be done under the long-term microgravity conditions available aboard the International Space Station.

We are studying the effects of modeled microgravity on erythroid cell surface receptors, receptor-initiated signal

Task Description:	transduction and on the assembly of the cytoskeleton in order to elucidate how these effects influence cell growth and differentiation. We are testing the following hypotheses. 1. Gravitational forces affect the interaction of cell surface erythropoietin receptors with erythropoietin. 2. These changes in receptor function are due, at least in part, to direct effects on receptor distribution, expression and/or protein-protein interactions. 3. Gravitational forces on erythroid cells also affect downstream signaling events both indirectly, due to effects on the erythropoietin receptor, and directly, through forces that modify signaling molecules within the cell. 4. During erythropoietin-induced cell growth and differentiation, gravitational forces modulate the assembly of the cytoskeleton, which is an integral part of terminal differentiation. The experiments are designed to elucidate gravitational effects on signal transduction, gene expression, and differentiation of red blood cell progenitors. Previous experience with our Epo sensitive model erythroid progenitor system and with techniques related to signal transduction and the cytoskeleton and the availability of all required materials and equipment assure that we will be able to achieve the specific aims of the study within three years. Preliminary data we have generated and previously published related work suggest that we will be able to identify and characterize significant effects of shear on signal transduction, gene expression, and differentiation in red blood cell progenitors. The information obtained through this study is likely to be applicable to the anemia associated with space flight and is also likely to be applicable to a wide range of other biologic effects of microgravity, since the processes we propose to study are basic to the functioning of many cell types.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The information obtained through this study is likely to be applicable to the anemia associated with space flight and is also likely to be applicable to a wide range of other biologic effects of microgravity, since the processes we propose to study are basic to the functioning of many cell types. The experiments will further the use of low shear earth-based bioreactors, such as the RWV, in carrying out modeled microgravity experiments on earth, the conclusions of which can be tested and validated on board the ISS. The results of these experiments will have impact on our understanding of erythropoietin receptor function, signal transduction and eukaryotic cell growth and differentiation. Additionally, they will also have important implications for space-based studies of cell and molecular biology. This work will advance the use of the space environment to increase scientific and medical knowledge and to enhance the quality of life on earth.
Task Progress:	We have developed a highly specific and sensitive assay for hemoglobin (Hb), the prime marker of erythroid differentiation, in fixed Rauscher murine erythroleukemia cells (R-cells) ideally suited to space-based research. Cultured R-cells were induced with 50 U Epo/mL or with 0.7% DMSO and were fixed either with 5.8% formalin or with 1.6% paraformaldehyde + 2.4% glutaraldehyde. After removal of fixative, the cells were incubated with tetramethyl benzidine and hydrogen peroxide to generate a chromophore (catalyzed by the pseudoperoxidase activity of Hb) measured spectrophotometrically. We found that Hb in R-cells could be measured with a high specificity and sensitivity even after up to four months of fixation. We observed that 1) the absorbance of induced cells was higher ($p<0.05$) than that of non-induced ones, 2) as few as 1 x 105 cells could generate a specific signal, 3) absorbance was directly proportional to cell number and to Hb content and 4) incubation temperature, fixative type and duration affected chromophore production. This method is directly applicable to studies of erythroid differentiation on orbit. Changes in cell surface receptor expression and function are a potential means by which cells might sense gravity and hydrodynamic shear (mechanosensing). We have studied the erythropoietin and interleukin-3 receptor (EpoR & IL-3R) on BaF3 cells grown either in low shear conditions (LSC) in the NASA Rotating Wall Vessel bioreactor (RWV) or in static culture conditions (SCC) in conventional tissue culture flasks at either low or high density (HD). Cell surface receptor expression and receptor-mediated endocytosis were quantified by 125I-Epo and 125I -IL-3 binding. We found that under LSC at HD, cell surface EpoR expression was reduced significantly ($p<0.05$) being only 50-60% that seen under SCC. Hydrodynamic shear appears to have a global effect on cell surface receptors since IL-3R also reduced under LSC. Northern blots revealed no differences in EpoR mRNA. However, studies of receptor-mediated endocytosis demonstrated an increase in EpoR internalization under LSC, suggesting the possibility of reduced recycling of EpoR back to the cell surface and presumably accounting for the reduced cell surface expression. Further studies suggested that this effect on the EpoR required the presence of Epo ligand occupying the EpoR. Culture vessels were not the cause of the differences. These results demonstrate a marked effect of hydrodynamic shear on EpoR expression and processing and may have implications for cytokine function in microgravity.
Bibliography Type:	Description: (Last Updated: 03/08/2019)
Articles in Peer-reviewed Journals	Chen C, Sytkowski AJ. "Apoptosis-linked gene-2 connects the Raf-1 and ASK1 signalings." Biochem Biophys Res Commun. 2005 Jul 22;333(1):51-7. PMID: 15925322 , Jul-2005
Articles in Peer-reviewed Journals	Xu K, Feldman L, Davis KL, Sytkowski AJ. "Erythropoietin and IL-3 receptor cell surface expression is decreased under conditions that model some aspects of microgravity." Gravit Space Biol Bull. 2005 Jun;18(2):111-2. PMID: 16044632 , Jun-2005